

FOREMOST AGDM, AB, Canada

WMO: 712250

Lat: 49.4833N

Lon: 111.4861W

Elev: 2890

StdP: 13.22

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.7	-12.8	-25.0	1.5	-18.3	-18.7	2.2	-12.6	32.3	38.8	29.0	29.8	8.2	150	0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.3	90.9	60.9	87.3	60.1	83.5	59.4	64.9	80.2	63.0	80.1	61.4	78.6	10.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	85.3	69.9	57.3	77.9	67.8	55.2	72.2	65.5	31.4	80.5	29.9	80.0	28.7	78.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	25.4	22.4	DB	-27.8	96.3	5.3	3.6	-31.6	98.8	-34.8	100.9	-37.8	102.9	-41.6	105.5	(4)
(5)			WB	-28.1	69.2	5.3	3.2	-31.9	71.4	-34.9	73.3	-37.9	75.0	-41.7	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	20.8	19.4	31.7	42.8	51.9	59.3	66.5	64.9	55.5	43.3	30.1	19.9		
	DBStd	20.11	15.42	15.38	13.29	8.05	8.09	5.96	5.59	6.48	8.55	9.80	13.72	14.75		
	HDD50	4472	905	856	571	239	74	4	0	1	39	248	602	933		
	HDD65	8502	1370	1276	1034	667	412	189	49	84	302	673	1048	1398		
	CDD50	1662	0	0	3	22	133	283	512	461	204	40	4	0		
	CDD65	215	0	0	0	0	5	18	96	79	17	0	0	0		
	CDH74	4242	0	0	1	14	172	403	1713	1467	448	24	0	0		
	CDH80	1602	0	0	0	1	37	105	700	599	159	1	0	0		
(14) Wind	WSAvg	11.2	12.7	11.1	12.2	12.1	11.4	10.3	9.1	9.2	10.2	11.4	12.0	11.9		
(15) Precipitation	PrecAvg	14.3	0.8	0.6	0.9	1.1	1.9	2.2	1.3	1.5	1.5	0.6	0.7	0.9		
	PrecMax	24.6	2.4	1.7	2.0	3.9	4.5	5.6	4.5	4.4	7.1	1.4	2.1	2.2		
	PrecMin	9.1	0.1	0.0	0.0	0.0	0.4	0.0	0.0	0.2	0.1	0.0	0.1	0.2		
	PrecStd	4.6	0.7	0.4	0.6	0.8	1.2	1.3	1.0	1.1	1.6	0.4	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	54.8	65.1	75.6	84.4	89.1	95.4	94.2	90.8	77.5	64.3	51.7		
		MCWB	42.2	40.9	47.2	51.7	57.5	62.8	62.2	60.5	58.7	54.4	47.4	40.0		
	2%	DB	46.2	47.2	59.3	69.5	79.0	82.6	91.4	90.6	84.8	70.0	56.5	45.6		
		MCWB	38.5	37.9	44.8	49.8	55.3	60.4	62.1	59.9	56.8	51.2	44.0	38.2		
	5%	DB	42.7	42.6	54.7	64.1	74.1	78.7	87.5	86.9	79.1	64.2	51.7	42.0		
		MCWB	36.3	35.5	42.2	46.7	53.1	59.1	61.7	58.9	55.4	48.7	41.9	35.6		
	10%	DB	39.9	38.9	49.7	59.0	69.2	74.4	83.6	82.7	73.3	59.1	47.2	38.9		
		MCWB	34.9	33.8	39.6	43.7	51.1	57.3	61.2	58.4	53.4	46.2	38.9	33.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	42.4	48.3	52.9	58.6	66.4	69.0	64.5	61.3	55.7	49.3	42.2		
		MCDB	50.8	51.5	62.2	71.8	79.4	79.6	83.3	81.4	78.6	73.5	61.0	49.2		
	2%	WB	39.2	38.4	45.5	49.9	56.2	63.1	66.1	62.6	58.5	52.3	45.1	38.5		
		MCDB	45.6	46.3	57.5	68.1	74.9	78.1	80.7	81.4	79.2	67.4	55.8	44.9		
	5%	WB	36.7	36.0	42.9	47.5	54.3	60.6	64.1	61.2	56.4	49.4	42.3	36.0		
		MCDB	41.9	41.7	53.7	62.3	70.9	75.4	80.2	80.3	76.5	62.3	50.8	41.4		
	10%	WB	35.0	33.9	39.9	44.9	52.3	58.5	62.5	59.8	54.3	47.0	39.4	33.9		
		MCDB	39.7	38.7	48.8	56.9	66.5	71.6	79.9	78.2	71.9	58.0	46.6	38.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.3	21.0	24.0	23.8	23.9	29.3	28.8	26.4	23.4	20.3	18.2		
		MCDBR	19.1	23.6	28.4	33.7	32.9	31.7	36.4	37.6	36.9	31.2	25.5	19.4		
		MCWBR	13.4	15.7	16.0	16.5	13.9	13.4	13.5	13.3	14.8	15.3	15.3	13.4		
	5% WB	MCDBR	18.3	22.2	27.3	31.3	28.7	28.7	30.5	30.6	33.9	29.2	24.6	19.1		
		MCWBR	13.4	15.7	16.0	15.9	12.9	13.3	13.6	12.0	14.2	14.9	15.4	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.267	0.287	0.335	0.353	0.347	0.348	0.368	0.328	0.289	0.261	0.251		
	taud		2.448	2.486	2.485	2.349	2.340	2.409	2.419	2.331	2.420	2.516	2.504	2.431		
	Ebn at Noon		259	286	297	289	287	289	287	275	277	273	258	246		
	Edh at Noon		19	23	28	36	38	36	35	36	30	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		410	725	1137	1567	1854	2060	2243	1800	1277	801	454	320		
	RadStd		35	57	81	107	139	121	115	114	103	77	28	27		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page